Comparative Evaluation of Real-Time Communication Protocols in a Real Vehicle Teleoperation Platform

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AMPL Autonomous Mobility and Perception Lab

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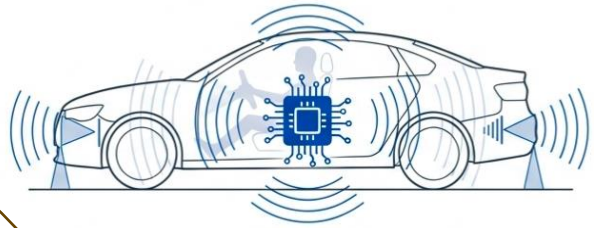
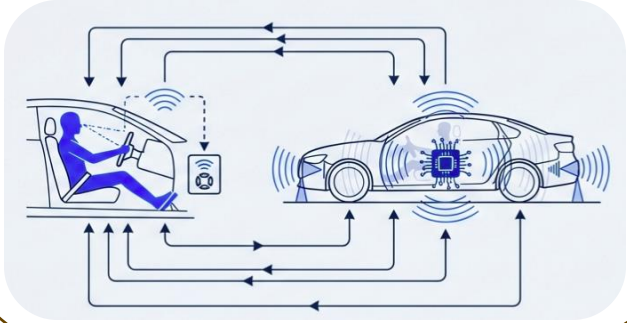
MOTIVATION AND CONTEXT

Intelligent Vehicles

Autonomous Vehicles



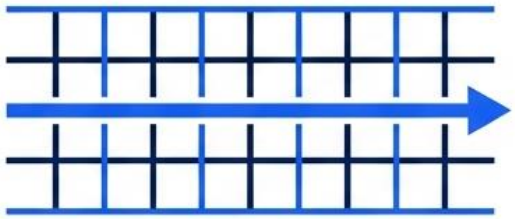
Teleoperation



Can face difficulties in unpredictable or extreme situations

PRIOR SOLUTIONS AND OBJECTIVE

Prior Solutions

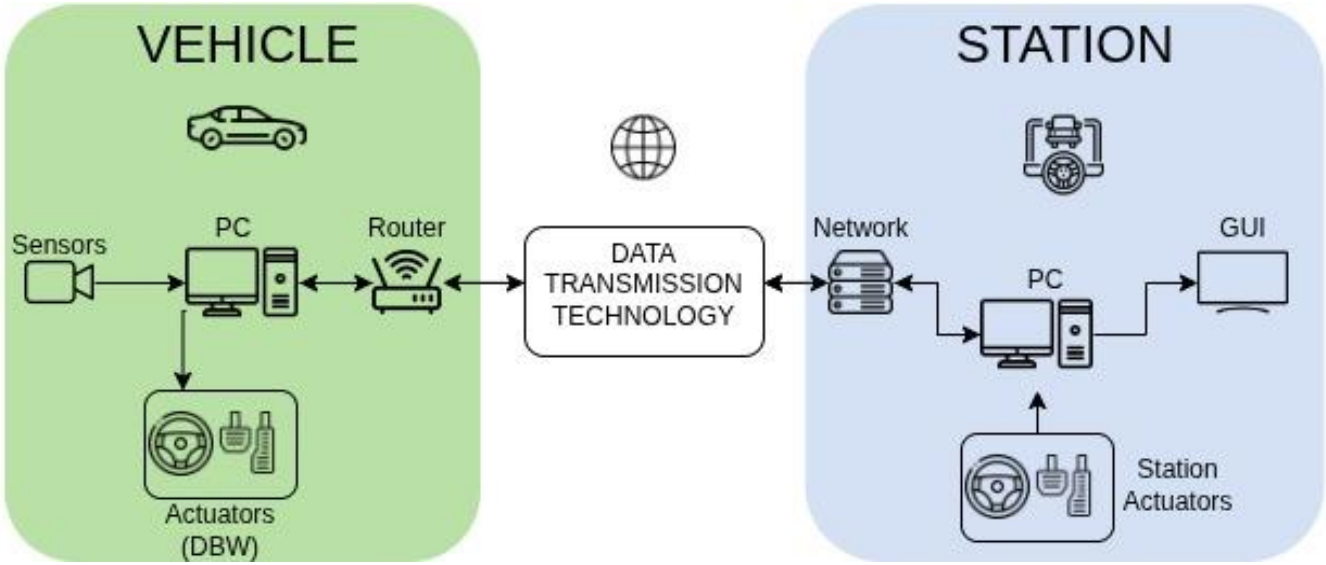


Limited Options
Only Simulations
Not teleoperation-focused

Objective

Multiple Technologies
Same conditions
Real-World Validation

SYSTEM OVERVIEW

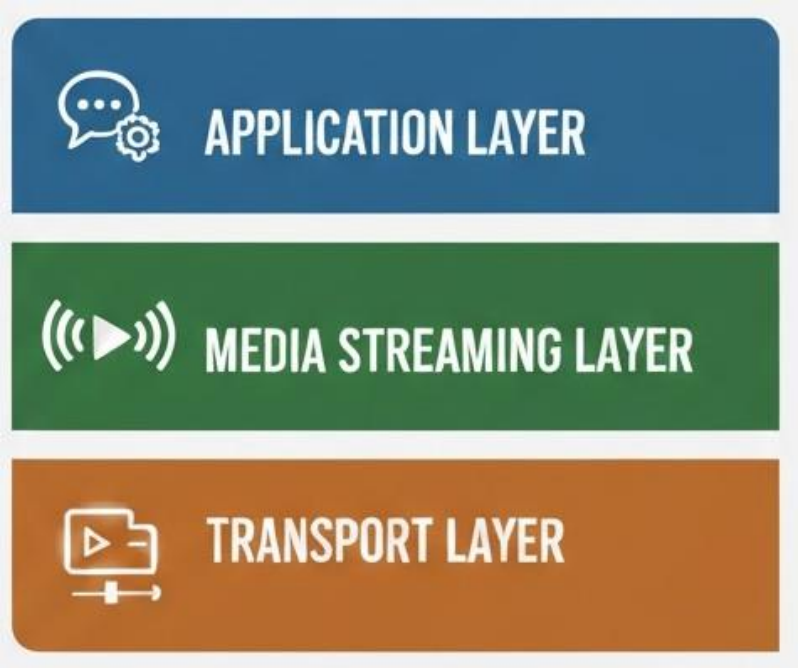


HARDWARE SETUP

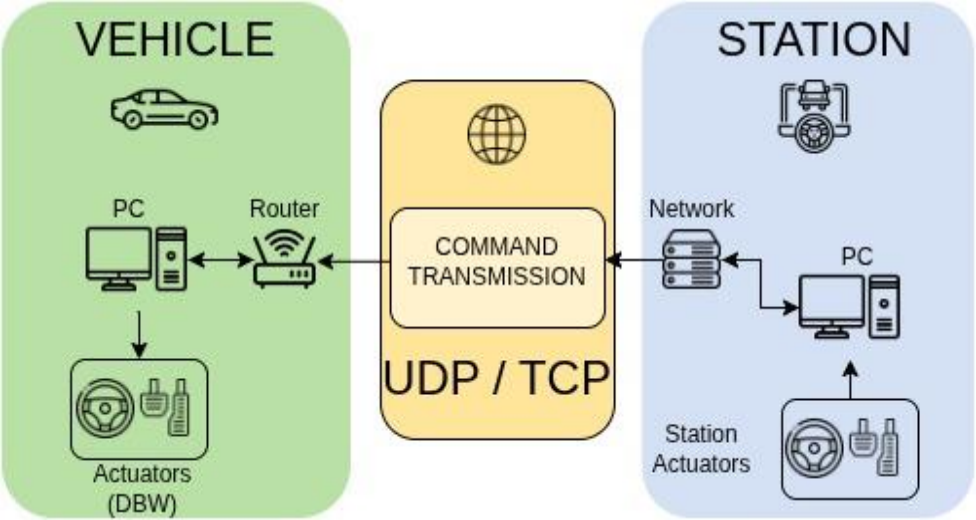
Vehicle

Teleoperation Station

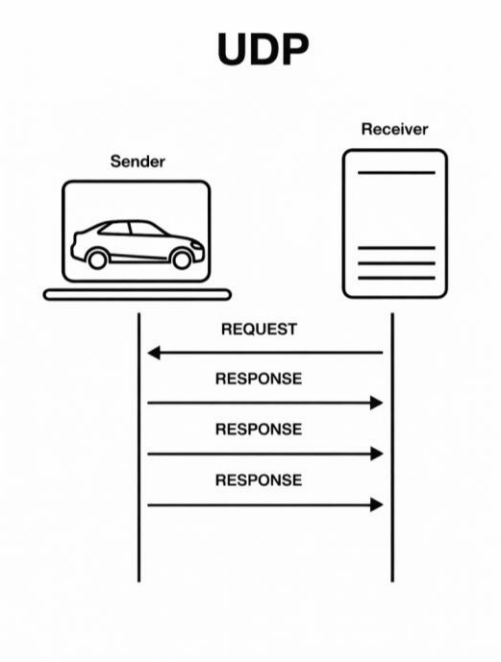
COMMUNICATION STACK ANALYSIS

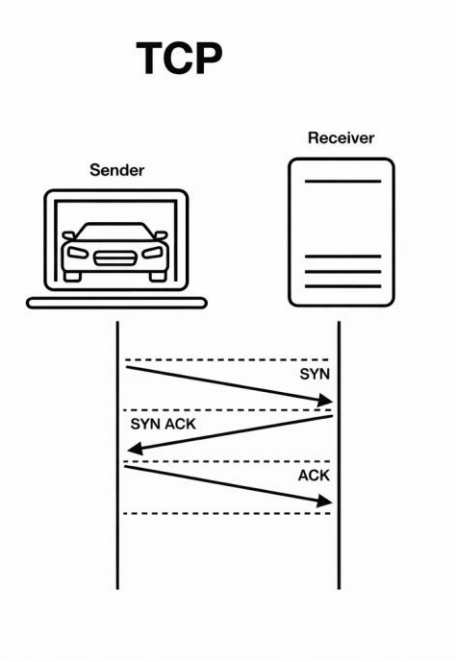


TRANSPORT LAYER ARCHITECTURE

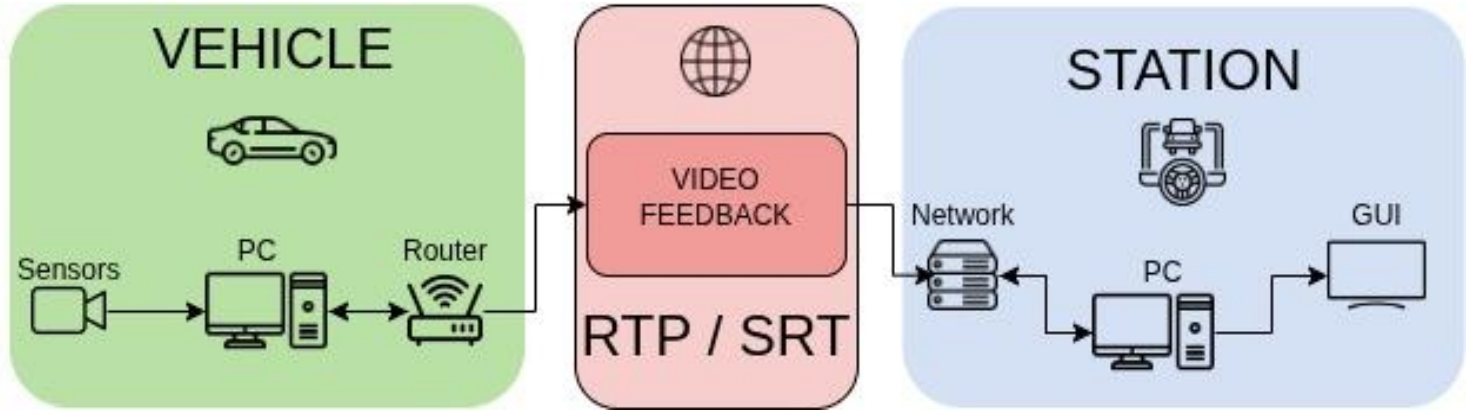


COMMAND TRANSMISSION: UDP / TCP

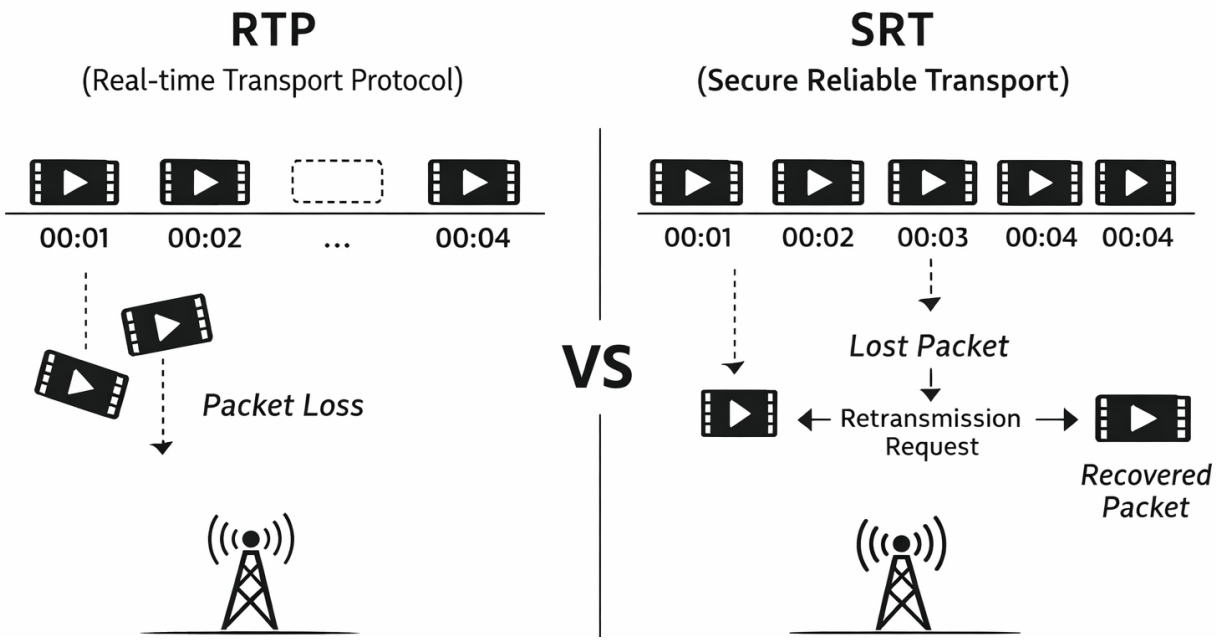




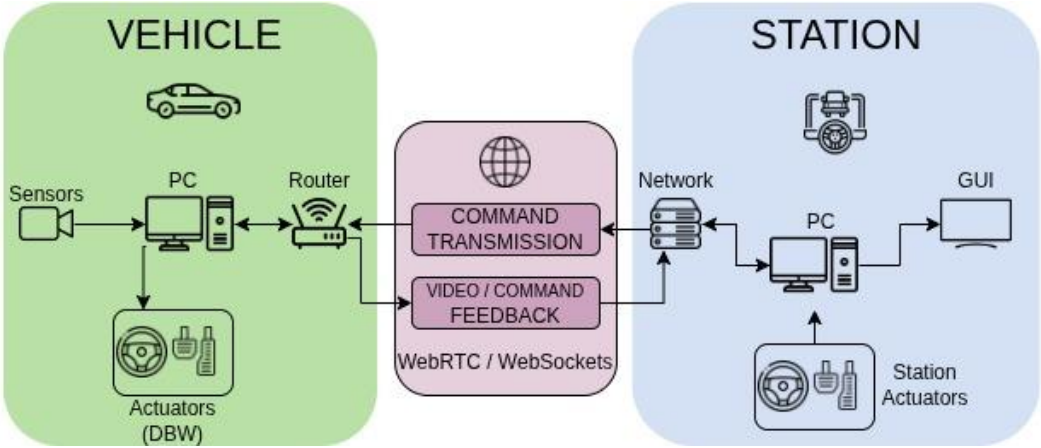
MEDIA STREAMING LAYER ARCHITECTURE



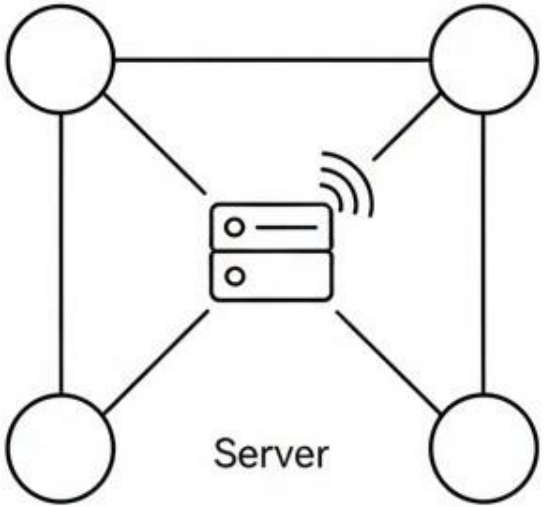
VIDEO FEEDBACK: RTP / SRT

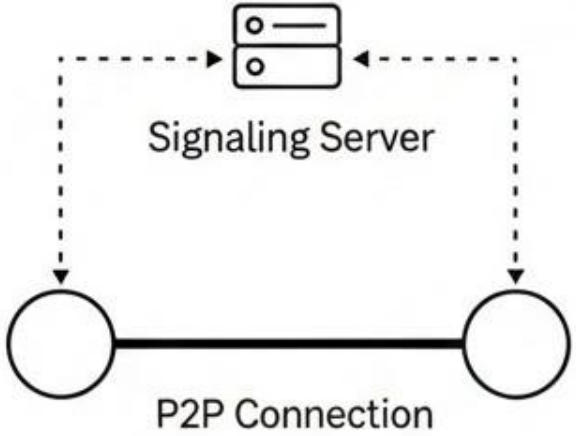


APPLICATION LAYER ARCHITECTURE



OPERATIONAL INTERFACE: WebSockets / WebRTC





EXPERIMENTAL RESULTS - THE TEST ROUTE

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EXPERIMENTAL RESULTS I - UDP vs TCP

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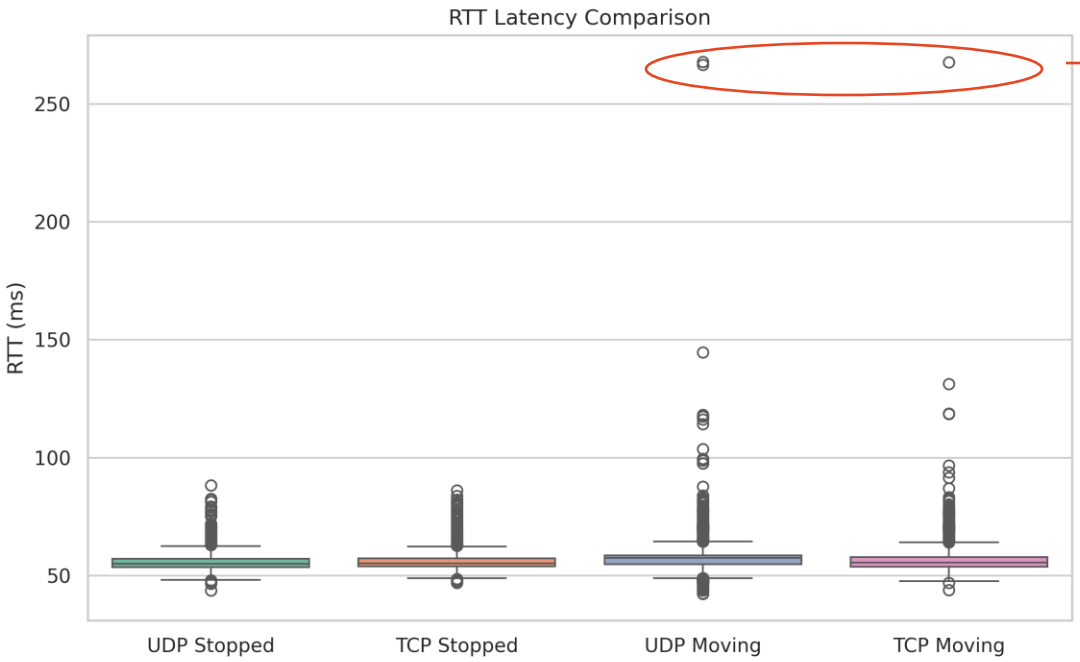

Signal Fading /
5G switching towers

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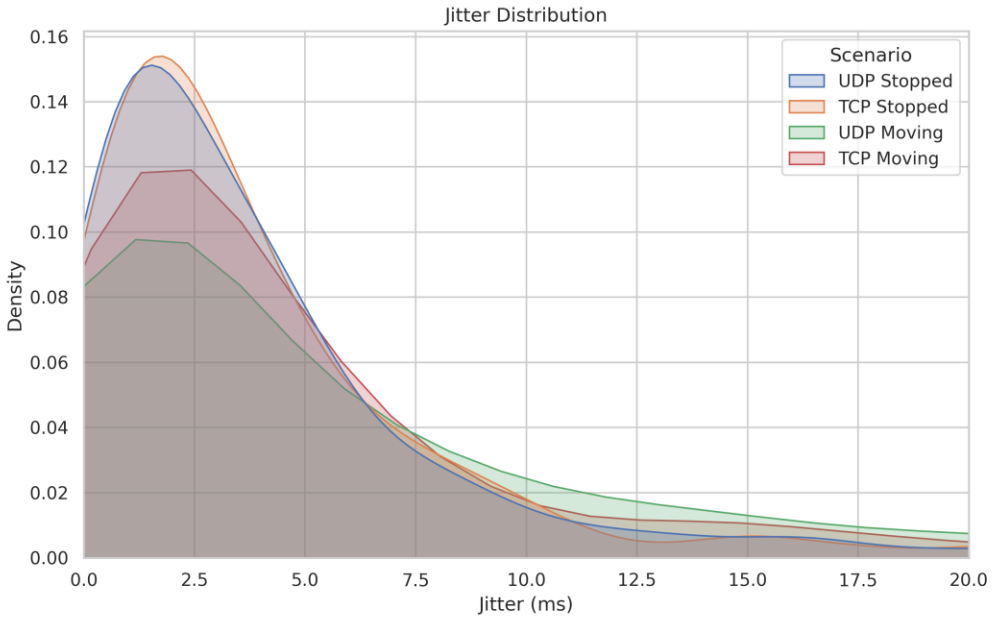
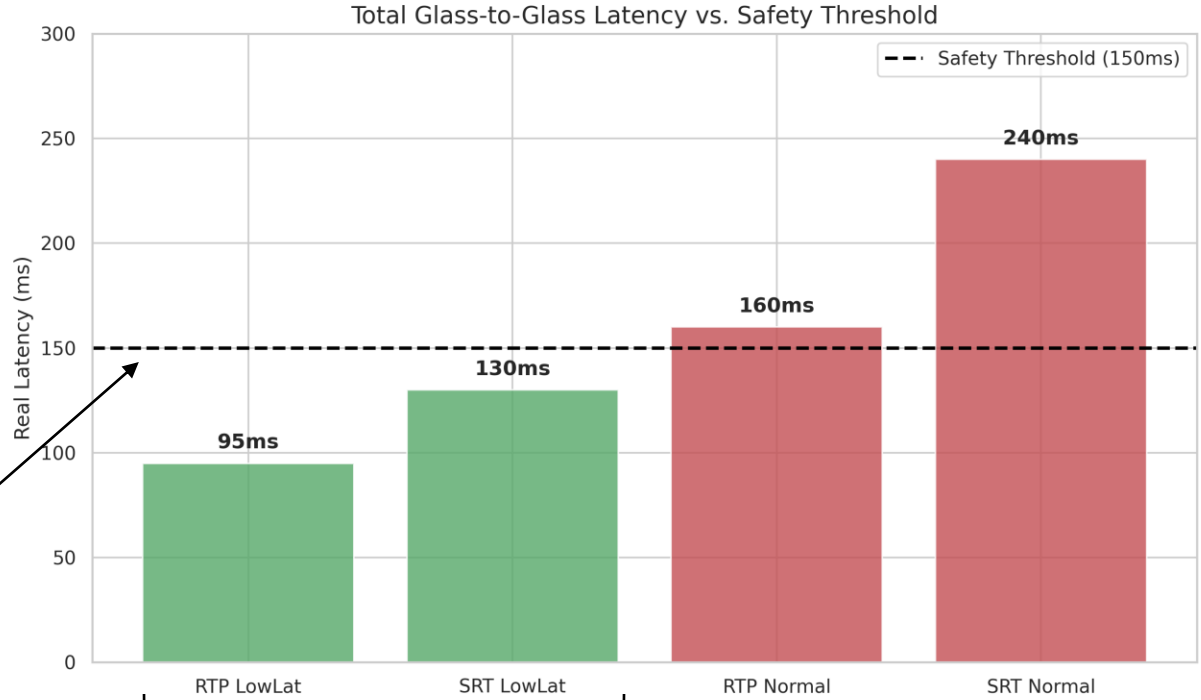
EXPERIMENTAL RESULTS I - UDP vs TCP



Jitter: Variation in latency over time

EXPERIMENTAL RESULTS II – RTP vs SRT

EXPERIMENTAL RESULTS II – RTP vs SRT



Max latency for safe reaction time

Optimized versions

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EXPERIMENTAL RESULTS III - WebSockets vs WebRTC

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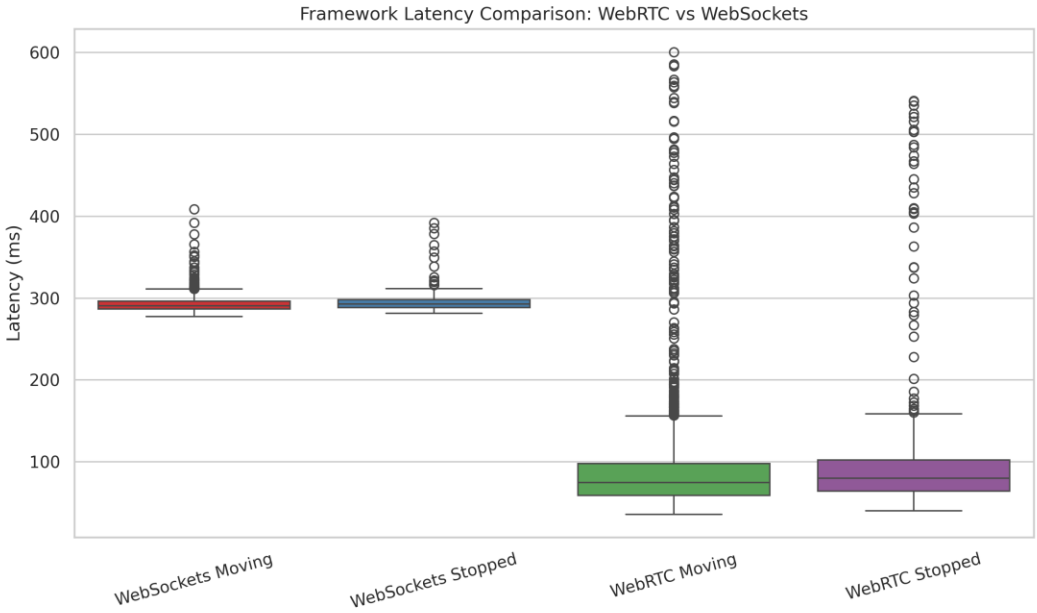
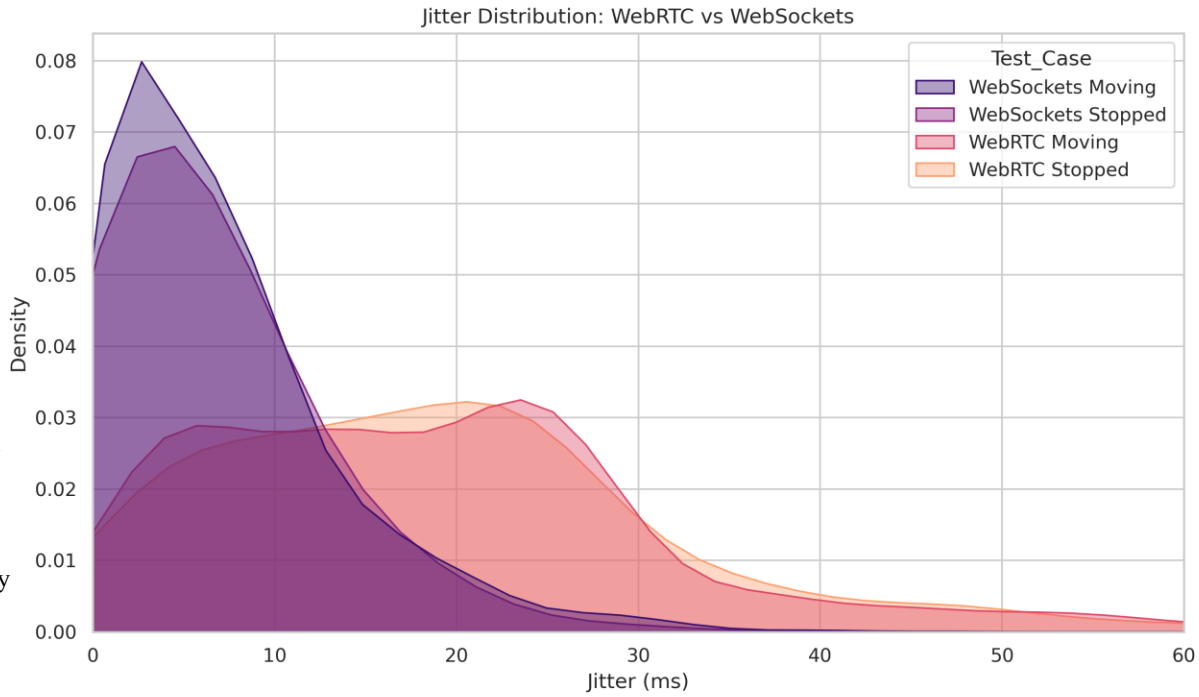


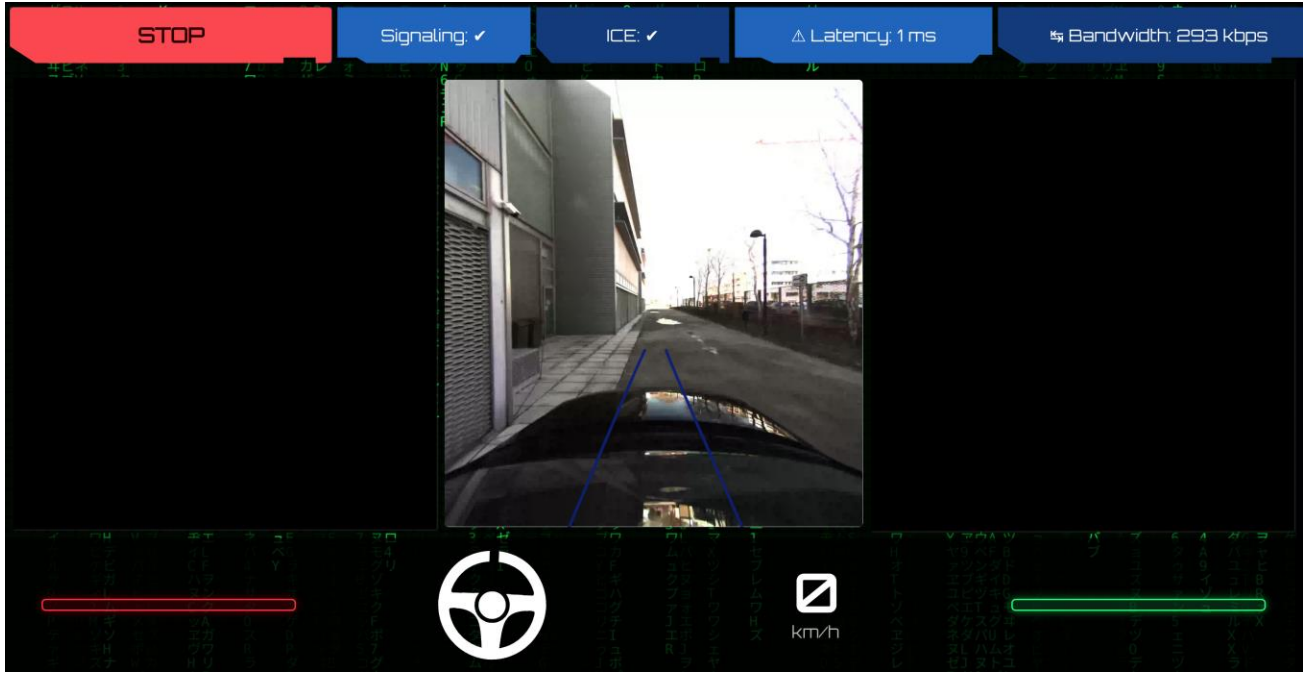
EXPERIMENTAL RESULTS III - WebSockets vs WebRTC



WebRTC's packet arrival time varies more

WebSockets look 'stable' but its latency is constantly high

CONCLUSIONS



THANK YOU

Questions?

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Disclosure of Interests. No competing interests

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